

Implicit Context Compression for Local Software Engineering Agents

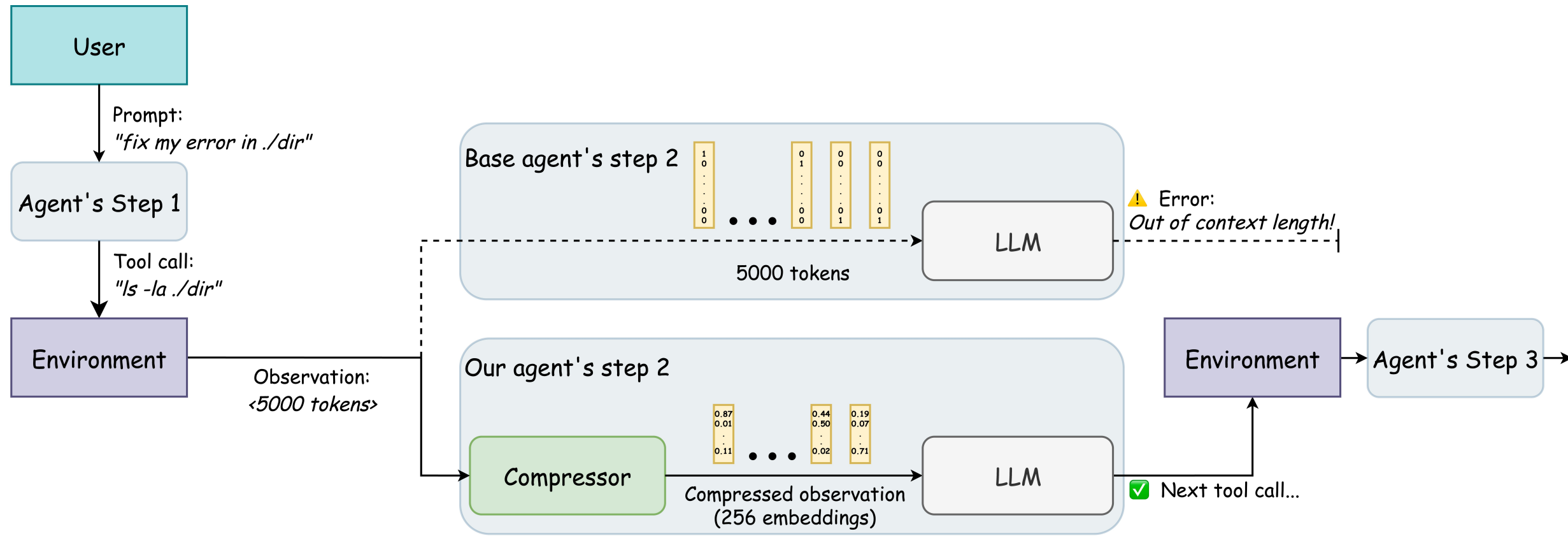
Kirill Gelvan*, Igor Slinko, Felix Steinbauer, Egor Bogomolov, Yaroslav Zharov

Motivation & Task Setup

Why and how?

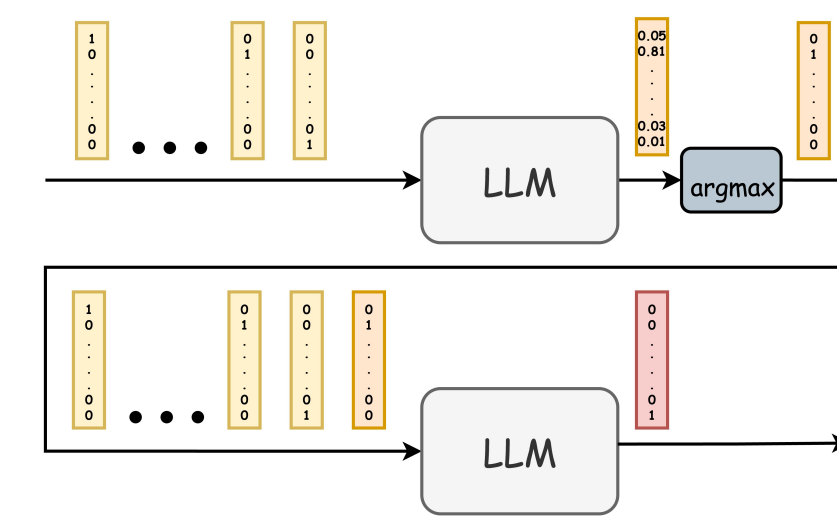
- LLMs have **limited context length**
- SWE Agents often call for tools with **unnecessarily long outputs**
- The **latent space** of embeddings is **much denser** than the discrete space of tokens
⇒
- Let's learn to compress environment observations

SWE-bench Verified[1] example of Implicit Context Compression



Results of Implicit Compression without Training

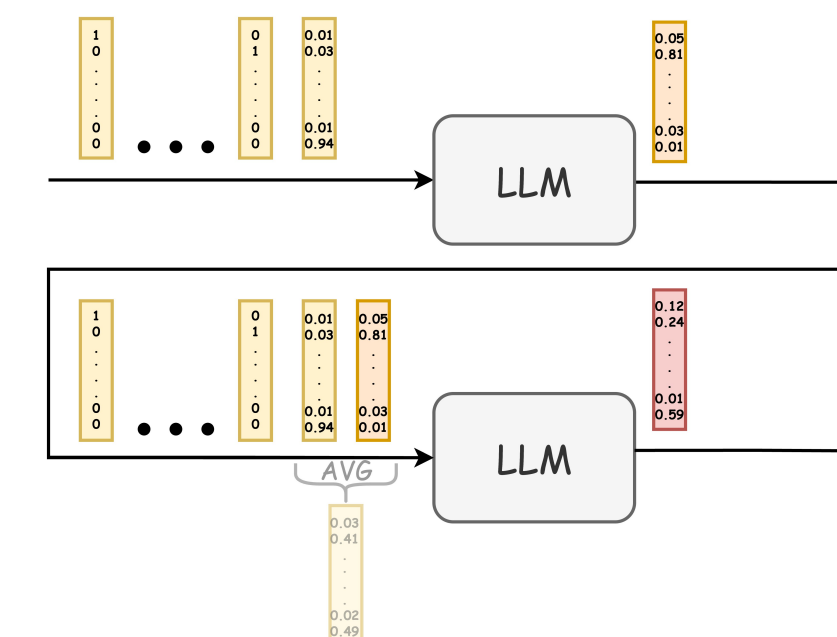
Hard Tokens (baseline)



SQuAD[2], context embed	F1
Baseline — hard tokens	0.71

Table: Hard tokens technique

Soft Embeddings Tokens (ours)



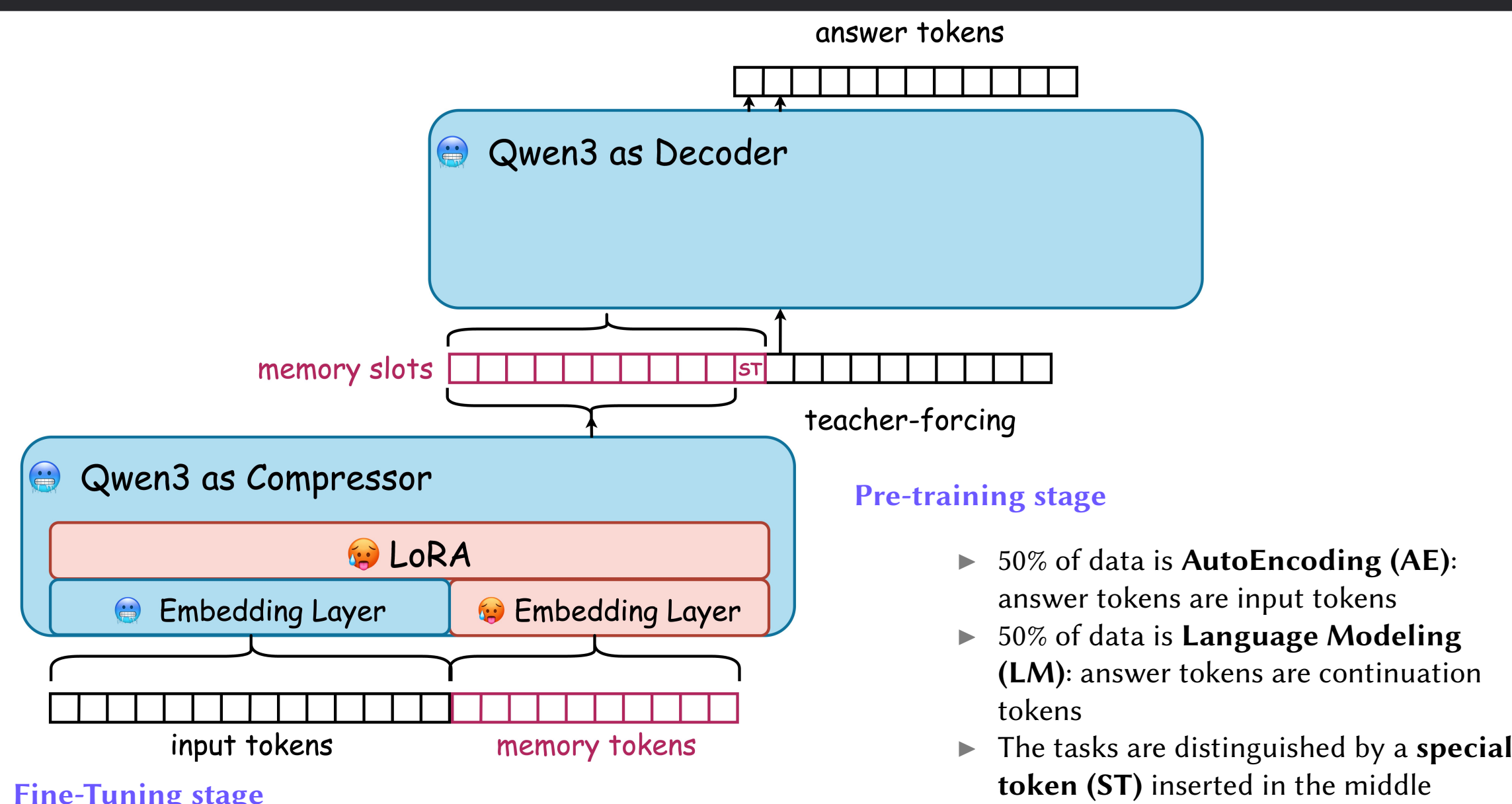
SQuAD[2], context embed	F1
Soft-embedded online	0.17
Soft-embedded online, avg x2	0.11
Soft-embedded "regenerate-llm", avg x2	0.16

Table: Soft-embedded techniques

Can we do it without training?

- No, we cannot!
- The scores decrease by more than 50% when using continuous representations (without avg)
- In order to get to the latent space, **we need training**

ICAE – In-Context AutoEncoder [3]



Fine-Tuning stage

- 100% of the data is from your own task. You may add a prompt after memory tokens as well

Note that only the "compressor" is trained, while **generation** relies on the **original Qwen** model!

Results of ICAE on General Texts & QA

Can we decompress texts after pre-training?

Dataset	Model	BLEU
PWC [3]	Mistral-7B	99.1
	Llama-2-7B	99.5
SQuAD (ours)	Qwen3-8B	98.1

Table: Text decomposition on PWC and SQuAD

Can we solve Question-Answering task after fine-tuning?

Model	Compression	Exact Match (%)	F1
Mistral-7B	x1	49	68
LoRA-FT Mistral-7B	x1	59	65
ICAE-FT Mistral-7B	x1.7 ± 0.7	69	73

Table: ICAE averaging on SQuAD

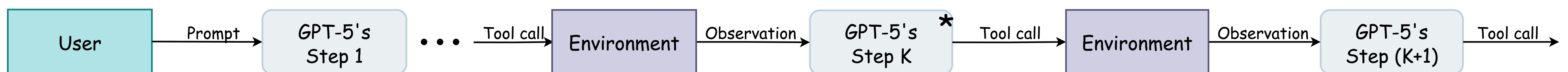
How can we interpret this?

- ICAE is able to decompress texts almost perfectly
- ICAE works with newer models and different datasets
- But does it work in an agentic setup?

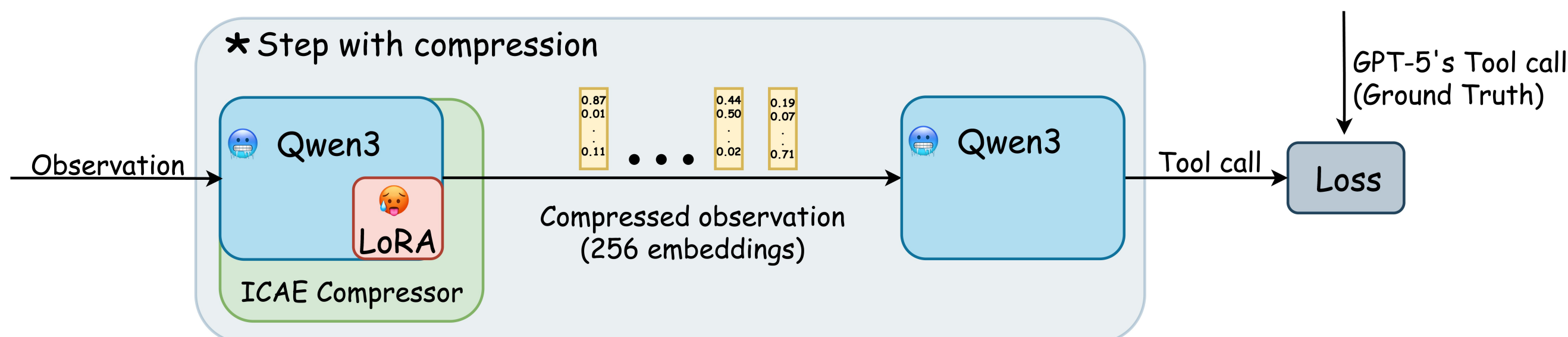
Training ICAE on SWE-bench

1 – Pre-train ICAE on general text corpora

2 – Get trajectories from a strong model (GPT-5)



3 – Fine-Tune Qwen3's LoRA on steps with compression:



Results

- Latency:** ICAE compression shows a theoretically **10% faster** mean tool-call generation time than vanilla Qwen3-8B
- Token-wise accuracy:** Qwen3-8B with and without compression perform **on par**
- Resolved on SWE-bench Verified:** The model with compression **resolves fewer than 50%** as many issues as the original.

Hypotheses

- Representation-behavior mismatch:** The ICAE encoder boosts token-level accuracy slightly, but perturbs decoder behavior for tool use, causing fewer end-to-end "resolved" completions
- Reconstruction quality falloff:** The model is unable to fully reconstruct code files. It is either not applicable to agentic setup or the pre-training should be longer
- Overfitting:** The low resolved rate with the higher accuracy suggests overfitting to labels rather than actual learning to solve SWE tasks

[1] Neil Chowdhury et al. Introducing SWE-bench Verified. 2024.

[2] Pranav Rajpurkar et al. "SQuAD: 100,000+ Questions for Machine Comprehension of Text".

[3] Tao Ge et al. "In-context Autoencoder for Context Compression in a Large Language Model".

